Interactive Visual Analytics for Brain Data Exploration

PHASE II — DESIGN, IMPLEMENTATION & VALIDATION PLAN

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1. Project Background

- Brain-related datasets can be high-dimensional, complex and difficult to interpret using raw tables or static charts.
- Phase-I report identifies interactive visual analytics as a way to explore brain data and make patterns easier to understand.
- The proposed platform combines data input, preprocessing, visual exploration and a user-friendly interface.
- The prototype in this package uses clearly labelled synthetic EEG-like data for demonstration—not real patient data.

2. Problem Statement

- Brain signal and neuroimaging data may contain many observations, channels, variables and time points.
- Static outputs make it harder to compare channels, inspect changes over time and explore selected portions of a dataset.
- A lightweight interactive environment can support exploratory analysis through filters, signal plots, summary statistics and heatmaps.
- The system is an exploration and learning aid; it is not intended to diagnose medical conditions.

3. Objectives from Phase I

- Load and explore brain-related datasets in a structured workflow.
- Provide interactive visualizations and filters so users can inspect selected data and parameters.
- Support interpretation through time-series plots, frequency-oriented views and heatmaps where appropriate.
- Use basic data analysis and, where useful, machine-learning methods to explore patterns or groups.
- Keep the interface understandable for students and researchers.

4. Scope & Boundaries

- Initial demonstration: tabular signal data with time and multiple channels.
- Core functions: upload CSV, preview records, choose a channel, set a time window, view a signal chart and summary metrics.
- Planned extensions: frequency analysis (FFT), correlation heatmap, modality-specific visualizations and reproducible export.
- MRI/fMRI/DTI/MEG integration requires modality-specific formats, libraries and validation; it is not claimed as complete.
- No clinical diagnosis, treatment recommendation or patient-level prediction.

5. Proposed System Architecture

- 1. Data input: CSV upload or included synthetic demonstration dataset.
- 2. Validation: check required time column, numeric channels, missing values and malformed rows.
- 3. Preprocessing: sort by time, select channels, handle missing values transparently.
- 4. Analytics: summary statistics, correlation and optional frequency-domain features.
- 5. Visualization layer: interactive time-series, channel comparison and heatmap.
- 6. User interface: controls, data preview, findings and exportable outputs.

6. Technology Stack

- Python: data processing and numerical calculations.
- Pandas / NumPy: tabular data handling and numeric operations.
- Plotly: interactive charts in the browser prototype.
- Streamlit (optional deployment route): rapid dashboard interface if a Python app is preferred.
- HTML, CSS and JavaScript version included as a self-contained browser demo; no installation required.
- GitHub/GitLab can be used for version control and evidence sharing.

7. Prototype Walkthrough

- Open index.html in a modern browser.
- Use the built-in synthetic dataset or upload a CSV containing a time column and numeric signal columns.
- Select the channel and adjust the displayed time range.
- Inspect the time-series chart, summary statistics and correlation heatmap.
- Download the currently loaded dataset as CSV.
- All sample observations are simulated and must not be reported as experimental clinical findings.

8. Validation & Testing Plan

- Input validation: test valid CSV, missing time column, empty file and non-numeric columns.
- Data quality: check missing values, duplicate timestamps and channel count.
- Chart validation: compare displayed points with source rows and verify time-window filtering.
- Statistics validation: cross-check mean, standard deviation and correlation on small hand-checkable samples.
- Usability: ask team members to complete upload, filter, interpret and export tasks; record issues.
- Performance: measure load/render time on a stated dataset size and record browser/device conditions.

9. Current Implementation Status

- Delivered in this package: browser-based prototype with synthetic sample data, CSV upload, channel selection, time-window filter, signal visualization, summary statistics, correlation heatmap and CSV export.
- The prototype is a demonstrator; it has not yet been validated on real EEG/MRI/fMRI/MEG datasets.
- No quantitative performance benchmarks or user-study results are claimed at this stage.
- Next work: run documented test cases, fix defects, add frequency analysis and test with a suitable public dataset.

10. Work Breakdown Structure

- Task 1 — Requirements and dataset selection: define supported input schema and evaluation questions.
- Task 2 — Data layer: parsing, validation, missing-value reporting and preprocessing.
- Task 3 — Visualization: signal chart, channel comparison and correlation heatmap.
- Task 4 — Interaction: filters, dataset upload, data preview and export.
- Task 5 — Validation: test cases, known-answer checks, usability review and performance measurements.
- Task 6 — Documentation: architecture diagram, screenshots, results table, limitations and final report.

11. Milestones & Evidence

- Milestone A: architecture and input schema approved by guide.
- Milestone B: core prototype runs locally and supports the sample dataset.
- Milestone C: CSV validation and visualization test cases completed.
- Milestone D: at least one suitable public dataset tested, with source and license documented.
- Milestone E: screenshots, test log, findings and updated report prepared.
- Evidence to share: source code repository, demo video, screenshots, architecture document and testing log.

12. Risks, Ethics & Limitations

- Different brain-data modalities use different formats and require separate processing workflows.
- Synthetic signals are useful for interface testing but cannot validate scientific or clinical usefulness.
- Public datasets must be checked for license, provenance, privacy and documentation.
- Visual patterns may be misleading if axes, filtering or missing-data handling are not clearly shown.
- The tool is for exploratory visualization only and must not be presented as a diagnostic system.

13. Expected Phase-II Deliverables

- A working exploratory prototype with documented input requirements.
- System architecture and data-flow diagram.
- Test-case table with expected and observed outcomes.
- Evidence package: screenshots, test log, dataset citation and short demonstration video.
- Updated technical report with implemented features, limitations and future scope.

14. Conclusion

- Phase II converts the Phase-I objectives into a testable design and an incremental prototype.
- The first implementation focuses on structured signal exploration and transparent visualization.
- Validation will be based on recorded test results—not assumptions or expected results.
- Future extensions depend on successful testing and suitable datasets.